

Thermo Homework # 3

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1.) Find the following properties using tables:

a.) Find T in $^{\circ}\text{F}$, v in $\frac{\text{ft}^3}{\text{lbm}}$, h in $\frac{\text{Btu}}{\text{lbm}}$ for water @ $p = 100 \frac{\text{lbf}}{\text{in}^2}$, $u = 500 \frac{\text{Btu}}{\text{lbm}}$

Using Table: @ $p = 100 \frac{\text{lbf}}{\text{in}^2}$

$$u_f = 298.19 \frac{\text{Btu}}{\text{lbm}}, \quad u_g = 1105.5 \frac{\text{Btu}}{\text{lbm}}, \quad h_f = 298.51 \frac{\text{Btu}}{\text{lbm}}, \quad h_g = 1187.5 \frac{\text{Btu}}{\text{lbm}}$$

$$v_f = 0.01774 \frac{\text{ft}^3}{\text{lbm}}, \quad v_g = 4.4327 \frac{\text{ft}^3}{\text{lbm}}, \quad u_1 = 500 \frac{\text{Btu}}{\text{lbm}}$$

$$u_1 = u_f + x u_g \Rightarrow 500 \frac{\text{Btu}}{\text{lbm}} = 298.19 \frac{\text{Btu}}{\text{lbm}} + x (1105.5 - 298.19) \frac{\text{Btu}}{\text{lbm}}$$

$$\Rightarrow x = 0.25$$

$$h = h_f + x (h_g - h_f) = 298.51 + 0.25 (1187.5 - 298.51) \frac{\text{Btu}}{\text{lbm}}$$

$$h = 520.75 \frac{\text{Btu}}{\text{lbm}}$$

$$v = v_f + x (v_g - v_f) = 0.01774 + 0.25 (4.4327 - 0.01774)$$

$$v = 1.121 \frac{\text{ft}^3}{\text{lbm}}$$

$$T_1 = T_{\text{sat}} @ 100 \frac{\text{lbf}}{\text{in}^2} = 327.81^{\circ}\text{F}$$

$$h = 520.75 \frac{\text{Btu}}{\text{lbm}}$$

$$v = 1.121 \frac{\text{ft}^3}{\text{lbm}}$$

$$T = 327.81^{\circ}\text{F}$$

Final Answer

1.b.) For ammonia @ $T = 40^\circ\text{F}$, $x = 0.5$, Determine P , v , and h using Table:

$$u_f = 85.98 \frac{\text{Btu}}{\text{lbm}} \quad u_g = 568.42 \frac{\text{Btu}}{\text{lbm}} \quad v_f = 0.02533 \frac{\text{ft}^3}{\text{lbm}} \quad v_g = 3.9664 \frac{\text{ft}^3}{\text{lbm}} \quad x = 0.5$$

$$v = v_f + x(v_g - v_f) \Rightarrow v = 0.02533 - 0.5(3.9664 - 0.02533)$$

$$v = 2 \frac{\text{ft}^3}{\text{lbm}}$$

$$u = u_f + x(u_g - u_f) \Rightarrow u = 85.98 - 0.5(568.42 - 85.98)$$

$$u = 327.2 \frac{\text{Btu}}{\text{lbm}}$$

$$P = P_{\text{sat}} @ 40^\circ\text{F} \quad P = 73.359 \frac{\text{lbf}}{\text{in}^2}$$

$$v = 2 \frac{\text{ft}^3}{\text{lbm}}$$

$$u = 327.2 \frac{\text{Btu}}{\text{lbm}}, \quad P = 73.359 \frac{\text{lbf}}{\text{in}^2}$$

Answer Part 1b.)

1c.) Water @ $P = 80 \text{ bar}$ and $v = 0.04034 \frac{\text{m}^3}{\text{kg}}$, find T , u , and h using Superheated Table:

$$P = 80 \text{ bar}, \quad v = 0.04034 \frac{\text{m}^3}{\text{kg}}$$

$$T = 480^\circ\text{C}$$

$$u = 3025.7 \frac{\text{kJ}}{\text{kg}}$$

$$h = 3348.4 \frac{\text{kJ}}{\text{kg}}$$

Answer part 1c.)

1d.) Refrigerant @ $T = -20^\circ\text{C}$, $h = 235.31 \frac{\text{kJ}}{\text{kg}}$. Find P , v , and u

using Table:

$$P = 1.3299 \text{ bar}$$

$$v = 0.1464 \frac{\text{m}^3}{\text{kg}}$$

$$u = 215.84 \frac{\text{kJ}}{\text{kg}}$$

Answer
Part 1d.)

2.) water is contained in a closed, rigid tank, initially a Saturated Vapor at 180°C is cooled to 100°C , find initial pressure in MPa and the final quality:

$$V_1 = V_2 \text{ (rigid tank)}$$

$$T_1 = 180^\circ\text{C}$$

Sat vapor table:

$$P_1 = P_{\text{sat}} @ 180^\circ\text{C} = 1002.8 \text{ kPa} \cdot \frac{0.001 \text{ MPa}}{1 \text{ kPa}} = \boxed{1.0028 \text{ MPa}}$$

$$v_1 = 0.19384 \frac{\text{m}^3}{\text{kg}}$$

$$v_1 = v_2 = v_f + x(v_g - v_f)$$

$$0.19384 \frac{\text{m}^3}{\text{kg}} = 0.001043 + x(1.6720 - 0.001043) \frac{\text{m}^3}{\text{kg}}$$

$$x = \boxed{0.11538}$$

3.) 2 kg of water vapor in a piston-cylinder assembly expands at constant pressure of 300 kPa (3.0 bar) from a saturated vapor to a volume of 2.064 m³. Determine T_{initial} , T_F , and work for the process;

$m = 2 \text{ kg}$ $P = 3 \text{ bar}$ $V = 2.064 \text{ m}^3$ State 1: Saturated Vapor

$P_1 = P_2$

v_1 (using table @ $P = 3 \text{ bar}$) = $v_g = 0.6058 \frac{\text{m}^3}{\text{kg}}$

$T_{\text{initial}} = 133.6^\circ \text{F}$

State 2:

$v_2 = \frac{V}{m}$ $V_1 = V_2$

$v_2 = \frac{2.064 \text{ m}^3}{2 \text{ kg}} = 1.032 \frac{\text{m}^3}{\text{kg}}$

@ $P = 3 \text{ bar}$, $v = 1.032 \frac{\text{m}^3}{\text{kg}}$

$T_F = 400^\circ \text{C}$

$V_{\text{initial}} = v_1 \cdot m$
 $= 0.6058 \frac{\text{m}^3}{\text{kg}} \cdot 2 \text{ kg} \quad \text{m}^3$

$W = P(V_2 - V_1)$

$V_{\text{initial}} = 1.2116 \text{ m}^3$

$W = 300 \text{ kPa}(2.064 - 1.2116) \text{ m}^3$

$W = 255.72 \text{ kJ}$

$T_i = 133.6^\circ \text{F}$

$T_F = 400^\circ \text{F}$

} Final
Answers

4.) Determine Initial Volume and Heat Transfer

$$P_1 = 10 \text{ bar} \quad T_1 = 60^\circ\text{C}$$

$$m = 1 \text{ kg} \quad W = 20 \text{ kJ}$$

$$P_2 = 20 \text{ bar}$$

$$T_2 = 30^\circ\text{C}$$

Using Table State 1

$$P_1 = 10 \text{ bar} = 1 \text{ MPa} \quad T_1 = 60^\circ\text{C}$$

$$v_1 = 0.023068 \frac{\text{m}^3}{\text{kg}}$$

$$u_1 = 270.32 \frac{\text{kJ}}{\text{kg}}$$

$$V_1 = m \cdot v_1 = 1 \text{ kg} \cdot 0.023068 \frac{\text{m}^3}{\text{kg}}$$

$$V_1 = 0.023068 \text{ m}^3$$

State 2:

$$P_2 = 20 \text{ bar} = 0.2 \text{ MPa} \quad T_2 = 30^\circ\text{C}$$

$$v_2 = 0.11847 \frac{\text{m}^3}{\text{kg}}$$

$$u_2 = 255.14 \frac{\text{kJ}}{\text{kg}}$$

$$Q = W + m(u_2 - u_1)$$

$$Q = 20 \text{ kJ} + 1 \text{ kg} (255.14 - 270.32) \frac{\text{kJ}}{\text{kg}}$$

$$Q = 4.82 \text{ kJ}$$

$$V_1 = 0.023068 \text{ m}^3$$

5.) Determine Initial Pressure and the heat transfer for the Process:

using Table A-4

$T_2 = 120^\circ\text{C}$, Saturated Vapor

$$v_2 = v_g = 0.89133 \frac{\text{m}^3}{\text{kg}}$$

$$u_2 = u_g = 2528.29 \frac{\text{kJ}}{\text{kg}}$$

$$v_1 = v_2 \text{ (rigid tank)}$$

$$= 0.89133 \frac{\text{m}^3}{\text{kg}}$$

$$\text{using table @ } T_1 = 70^\circ\text{C} \quad v_1 = 0.89133 \frac{\text{m}^3}{\text{kg}}$$

$$u_1 = u_f \left(\frac{v_1 - v_f}{v_g - v_f} \right) (u_g - u_f)$$

$$\Rightarrow 293.04 + \left(\frac{0.89133 - 0.001023}{5.0396 - 0.001023} \right) (2468.9 - 293.04)$$

$$u_1 = 677.51 \frac{\text{kJ}}{\text{kg}}$$

$$P_1 = P_{\text{sat}}$$

$$P_1 = 31.202 \text{ kPa}$$

Final
Answer

$$m = 1.5 \text{ kg}$$

$$Q = m(u_2 - u_1)$$

$$Q = 1.5(2528.29 - 677.51)$$

$$Q = 2776.17 \text{ kJ}$$

Final
Answer

6.) Determine the Volume of the tank and the fraction of total Volume occupied by Saturated Vapor,

$$P = 10 \text{ bar}$$

$$m_{\text{tank}} = 25 \text{ kg}$$

$$X = 0.6$$

Using Table: @ $P = 10 \text{ bar}$

$$v_f = 0.8352 \cdot 10^{-3} \frac{\text{m}^3}{\text{kg}} \quad v_g = 0.0236 \frac{\text{m}^3}{\text{kg}}$$

$$v = v_f + x(v_g - v_f)$$

$$v = 0.8352 \cdot 10^{-3} + 0.6(0.0236 - 0.8352 \cdot 10^{-3})$$

$$v = 0.01449 \frac{\text{m}^3}{\text{kg}}$$

$$X = \frac{m_g}{m_g + m_L}$$

$$0.6 = \frac{m_g}{m_g + 25} \Rightarrow m_g = 37.5 \text{ kg}$$

$$v = \frac{V}{m} \Rightarrow V = v \cdot m$$

$$V_g = 0.0236 \frac{\text{m}^3}{\text{kg}} \cdot 37.5 \text{ kg} = 0.885 \text{ m}^3$$

$$V_f = 25 \text{ kg} \cdot 0.8352 \cdot 10^{-3} \frac{\text{m}^3}{\text{kg}} = 0.02088 \text{ m}^3$$

$$V = m \cdot v$$

$$V = (25 + 37.5) \text{ kg} \cdot 0.01449 \frac{\text{m}^3}{\text{kg}} = 0.9056 \text{ m}^3$$

$$X_{\text{vapor}} = \frac{V_g}{V} = \frac{0.885}{0.9056} \cdot 100\%$$

$$X_{\text{vapor}} = 97.725\%$$

$$V = 0.9056 \text{ m}^3$$

Final Answer

7.) Determine R , initial Volume, and final Volume

$$m = 0.2 \text{ g} \cdot \frac{1 \text{ kg}}{1000 \text{ g}} = 0.0002 \text{ kg}$$

$$T_1 = 27^\circ\text{C} + 273$$

$$P_2 = 0.8 \text{ bar}$$

$$T_1 = 300 \text{ K}$$

$$T_2 = 17^\circ\text{C} + 273 = 290 \text{ K}$$

$$\bar{R} = 8.314 \frac{\text{kJ}}{\text{kmole K}}$$

$$R = \frac{\bar{R}}{M} = \frac{8.314}{4} = 2.0785 \frac{\text{kJ}}{\text{kg K}} \cdot \frac{1}{\frac{\text{kg}}{\text{kmole}}}$$

$$R = 2.0785 \frac{\text{kJ}}{\text{kg K}} \cdot \frac{1000 \text{ J}}{1 \text{ kJ}} \cdot \frac{1 \text{ kg}}{1000 \text{ g}} = 2.0785 \frac{\text{J}}{\text{g K}}$$

$$P_1 V_1 = m R T_1$$

$$V_1 = \frac{m R T_1}{P_1}$$

$$V_1 = \frac{0.2 \cdot 10^{-3} \cdot 2.0785 \cdot 300}{1.3 \cdot 10^2}$$

$$V_1 = 0.959 \cdot 10^{-3} \text{ m}^3 \cdot \frac{1000000 \text{ cm}^3}{1 \text{ m}^3} = 959 \text{ cm}^3$$

$$P_2 V_2 = m R T_2$$

$$V_2 = \frac{m R T_2}{P_2}$$

$$V_2 = \frac{0.2 \cdot 10^{-3} \cdot 2.0785 \cdot 290}{0.8 \cdot 10^2}$$

$$V_2 = 1.506 \cdot 10^{-3} \text{ m}^3 = 1506 \text{ cm}^3$$

$$V_1 = 959 \text{ cm}^3$$

$$V_2 = 1506 \text{ cm}^3$$

Final
Answer

$$R = 2.0785 \frac{\text{J}}{\text{g K}}$$